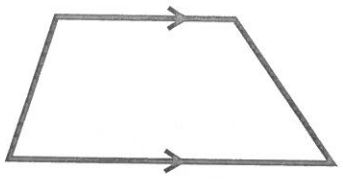
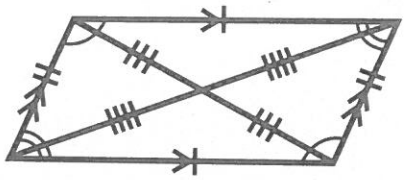
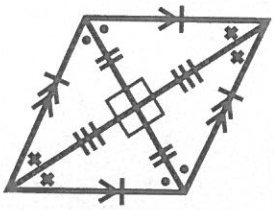
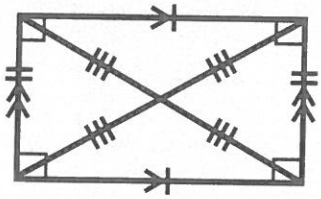
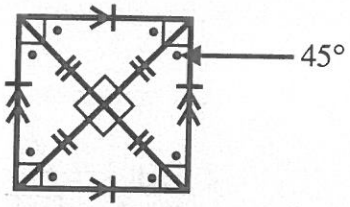
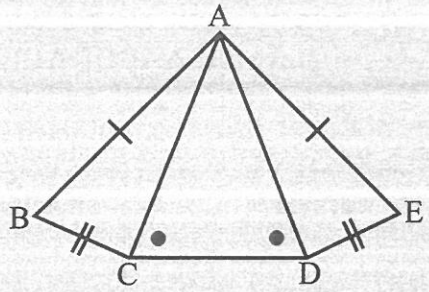
APPLICATIONS - QUADRILATERALS

The following table shows the properties of the different quadrilaterals studied in earlier grades:

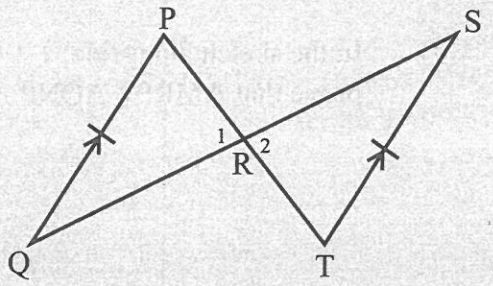
Quadrilateral	Properties
Trapezium 	• One pair of opposite sides parallel
Parallelogram 	• Both pairs of opposite sides parallel • Both pairs of opposite sides equal • Both pairs of opposite angles equal • Diagonals bisect each other
Rhombus 	• All four sides equal • Diagonals are perpendicular to each other • Diagonals bisect the angles (In addition to this, the rhombus also inherits all the properties of the **parallelogram**)
Rectangle 	• All four angles are 90° • Diagonals are equal in length (In addition to this, the rectangle also inherits all the properties of the **parallelogram**)
Square 	In a square, all four sides are equal and all four angles are 90°. (The square inherits ALL the properties of BOTH the **rhombus** and **rectangle**, and therefore also those of the **parallelogram**.)
Kite 	• Two pairs of adjacent sides equal • Diagonals are perpendicular to each other • The main diagonal bisects the other diagonal • The main diagonal bisects angles • The angles opposite the main diagonal are equal

In the next example, we will combine these properties with the distance, midpoint and gradient formulae to answer questions about a given quadrilateral:

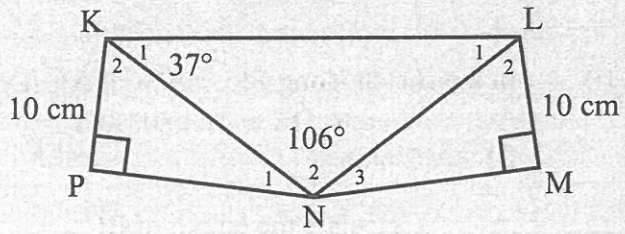
(2) In the sketch alongside, prove that $\Delta ABC \equiv \Delta AED$.



(3) In the sketch alongside, prove that $\Delta PQR \equiv \Delta STR$.



(4) In the sketch alongside, prove that $\Delta KPN \equiv \Delta LMN$.



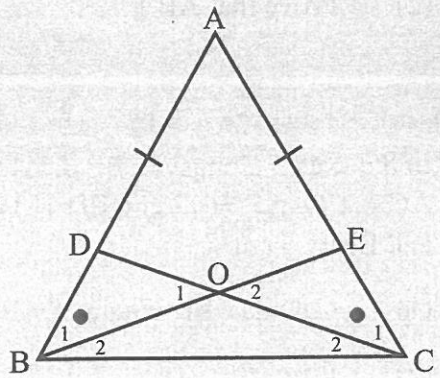
(g) In the sketch alongside, $AD = AE$ and $\hat{B}_1 = \hat{C}_1$.

Prove that

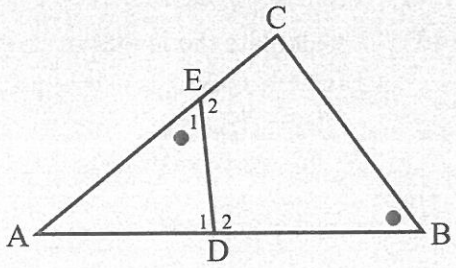
(1) $AB = AC$

(2) $BO = OC$

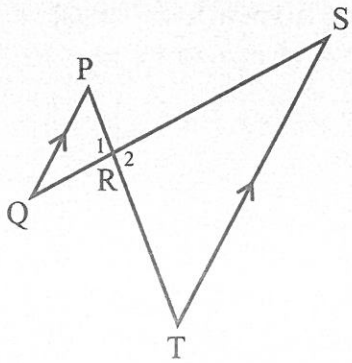
(3) $\Delta DOB \equiv \Delta EOC$



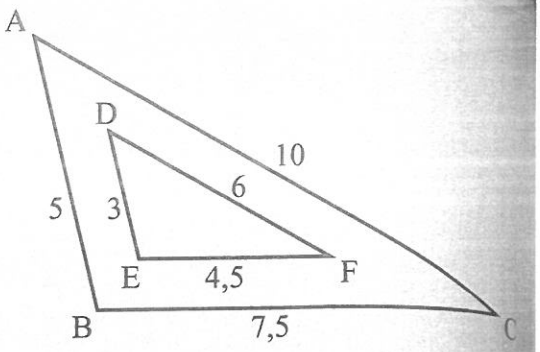
(h) (1) In the sketch alongside, prove that $\Delta ABC \,|||\, \Delta AED$.



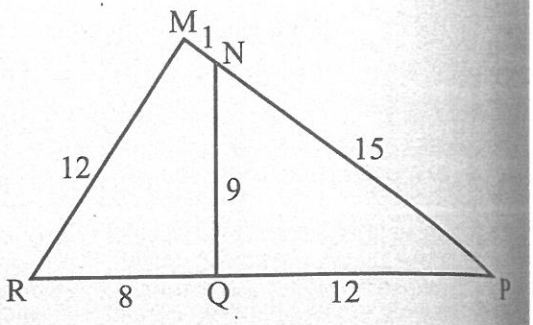
(2) In the sketch alongside, prove that $\Delta PQR \,|||\, \Delta TSR$.



(3) In the sketch alongside, prove that $\Delta ABC \,|||\, \Delta DEF$.



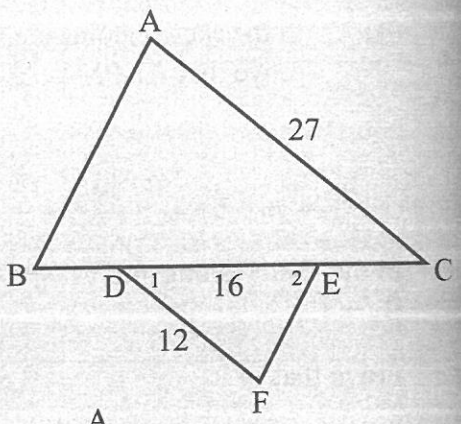
(4) In the sketch alongside, prove that $\Delta MRP \,|||\, \Delta QNP$.



(i) In the sketch alongside, $\Delta ABC \,|||\, \Delta FED$. AC = 27 units, DE = 16 units and DF = 12 units.

(1) Calculate the length of BC.

(2) Prove that AB ∥ EF.

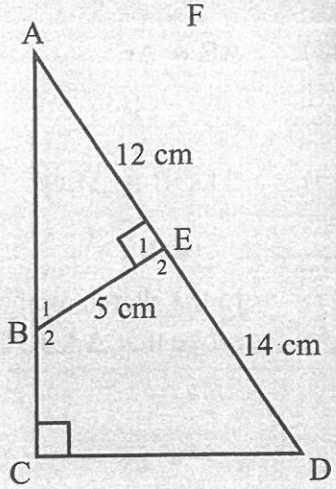


(j) In the sketch alongside, BE ⊥ AD and AC ⊥ CD. AE = 12 cm, ED = 14 cm and BE = 5 cm.

(1) Calculate the length of AB.

(2) Prove that $\Delta ABE \,|||\, \Delta ADC$.

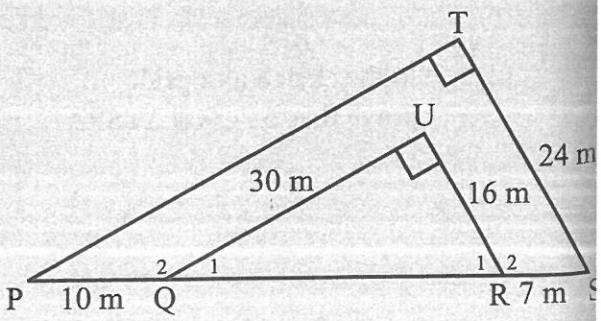
(3) Calculate the length of
(i) CD
(ii) BC



(k) In the sketch alongside, $\hat{T} = \hat{U} = 90°$. PQ = 10 m, RS = 7 m, QU = 30 m, UR = 16 m, and TS = 24 m.

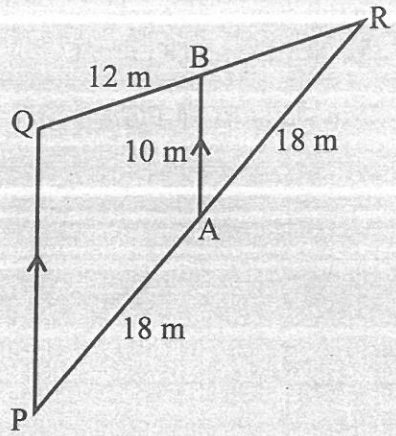
(1) Calculate the length of
(i) QR
(ii) PT

(2) Prove that PT ∥ QU.



(b) In the sketch alongside, PQ ∥ AB.
PA = 18 m, AR = 18 m, AB = 10 m and QB = 12 m.

Determine the length of
(1) BR
(2) PQ

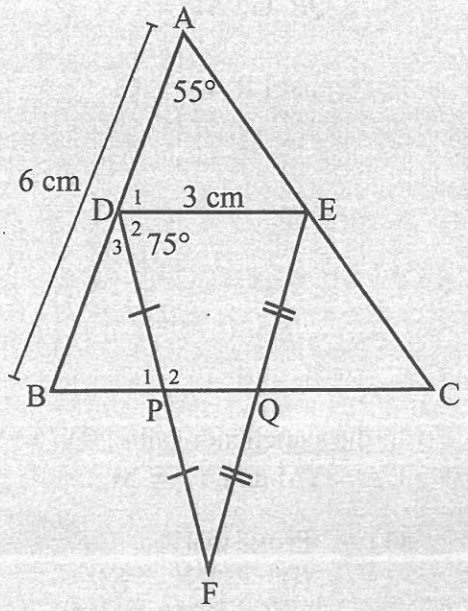


(c) In the sketch alongside, DP = PF and EQ = QF. AB = 6 cm and DE = 3 cm.
$\hat{A} = 55°$ and $\hat{D}_2 = 75°$.

(1) Determine the length of PQ.

(2) Determine the size of $\hat{P}_1$.

(3) If AE = EC, determine
(i) the length of AD.
(ii) the length of BC.
(iii) the size of $\hat{B}$.

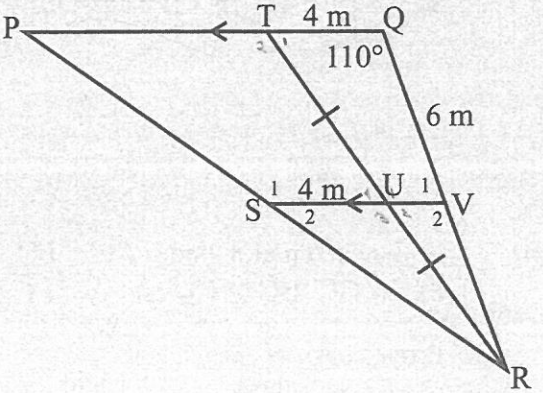


(d) In the sketch alongside, PQ ∥ SV and TU = UR. QV = 6 m, TQ = SU = 4 m and $\hat{Q} = 110°$.

(1) Determine the length of
(i) QR
(ii) UV

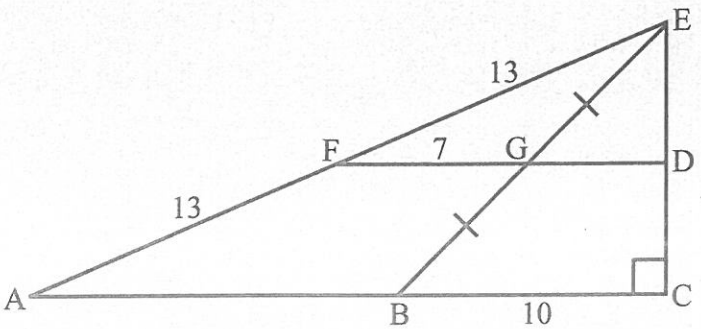
(2) Show that ΔPQR is isosceles.

(3) Calculate the size of $\hat{S}_2$.



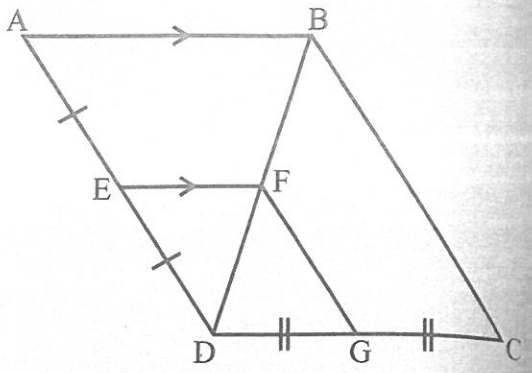
(e) In the sketch alongside, $\hat{C} = 90°$ and BG = GE. AF = FE = 13 units, BC = 10 units and FG = 7 units.

Determine the length of
(1) AB
(2) EC
(3) ED



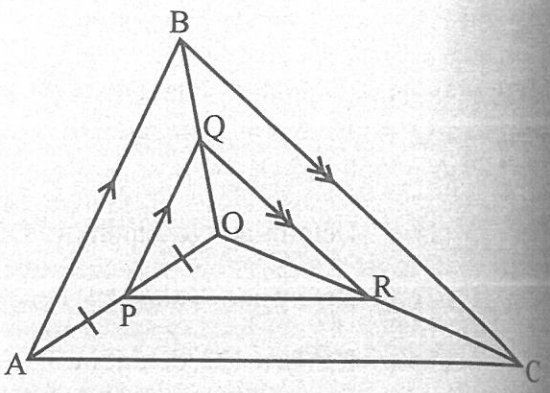
(f) In the sketch alongside, AB ∥ EF, AE = ED and DG = GC.

Prove that BC ∥ FG.



(g) In the sketch alongside, AB ∥ PQ, BC ∥ QR and AP = PO.

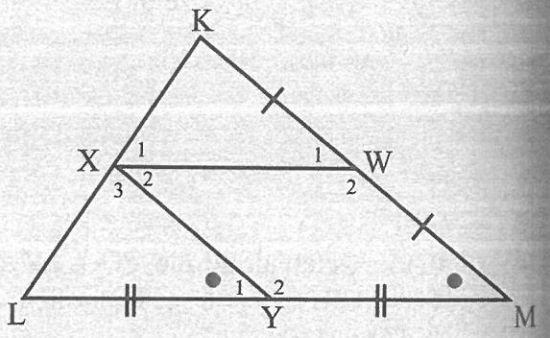
Prove that $PR = \frac{1}{2}AC$.



(h) In the sketch alongside, KW = WM, LY = YM and $\hat{Y}_1 = \hat{M}$.

(1) Prove that
- (i) KX = XL
- (ii) XW = YM

(2) If $\hat{Y}_1 = 40°$, calculate the size of $\hat{W}_2$.

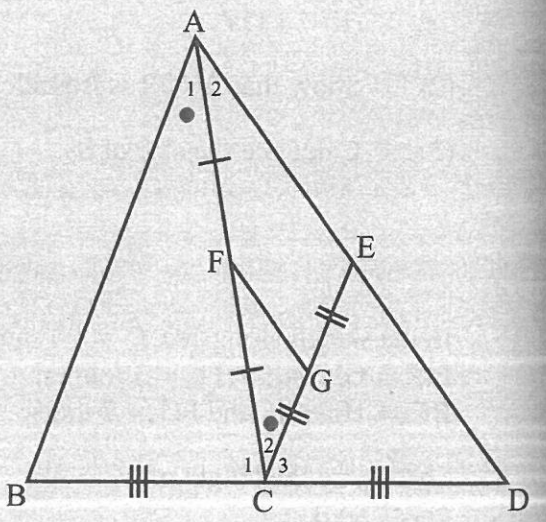


(i) In the sketch alongside, AF = FC, EG = GC, BC = CD and $\hat{A}_1 = \hat{C}_2$.

Prove that

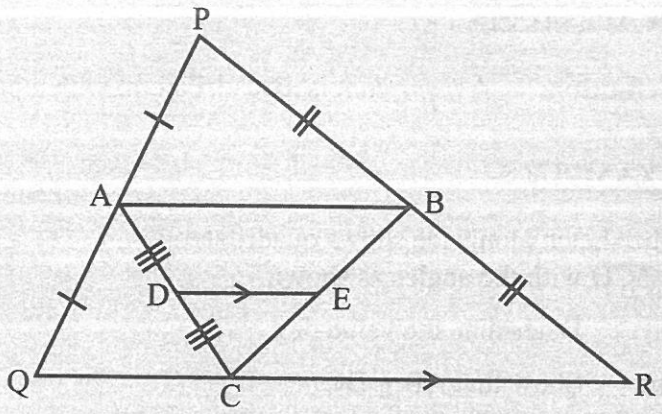
(1) AE = ED

(2) $FG = \frac{1}{4}AD$



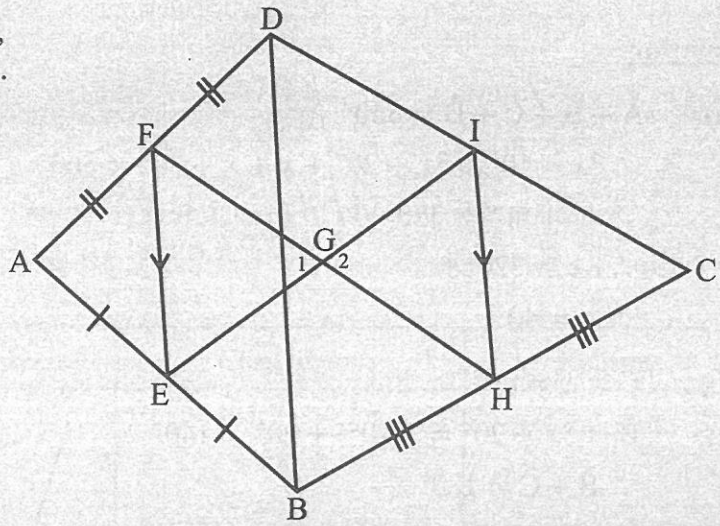
(j) In the sketch alongside, PA = AQ, PB = BR and AD = DC. DE ∥ QR.

Prove that
(1) BE = EC
(2) QR = 4DE



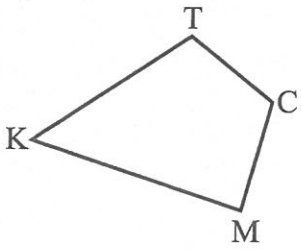
(k)* In the sketch alongside, AE = EB, AF = FD, BH = HC and FE ∥ IH.

Prove that
(1) DI = IC
(2) FG = GH



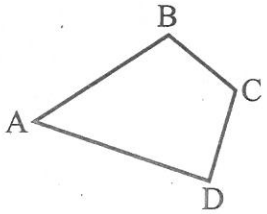
QUADRILATERALS

A *quadrilateral* is a flat shape enclosed by four straight line segments. A quadrilateral is always named by "going around the shape":



The quadrilateral alongside can be named KTCM or MCTK for example, but **not** KTMC.

The sum of the interior angles of any quadrilateral is 360°:

Theorem	Information given	Conclusion	Reason
The sum of the interior angles of a quadrilateral is 360°.		$\hat{A} + \hat{B} + \hat{C} + \hat{D} = 360°$	∠s of quad